

BAAVANSH REDDY GUNDLAPALLI

New Brunswick, NJ | (732) 213-6744 | baavanshreddy@gmail.com

linkedin.com/in/baavansh-reddy-gundlapalli | github.com/BaavanshReddy | baavansh-portfolio.vercel.app

SUMMARY

Backend and AI systems engineer with four years of engineering experience across production REST APIs, LLM and ML integration, and IoT platforms. Core stack of Python, Java, and C, with work ranging from authentication systems and data pipelines to compilers, CPU simulators, and operating systems. Author of AgentMemry, an open-source memory library for AI agents published on PyPI, and co-author of a retrieval evaluation study benchmarking BM25 against RAG. Personal portfolio hosts an AI assistant recruiters can chat with about this work.

EDUCATION

Rutgers University, New Brunswick, NJ

May 2026

B.S. in Computer Science, *Magna Cum Laude* | GPA: 3.76 / 4.0 | Dean's List: Fall '24, Spring '25, Fall '25, Spring '26

Relevant Coursework: Data Structures & Algorithms, Computer Architecture, Operating Systems, Compilers, Software Engineering, Databases, Artificial Intelligence

TECHNICAL SKILLS

Languages: Python, Java, C, C++, JavaScript, TypeScript, SQL

Frameworks & Web: FastAPI, Node.js, Express, Spring Boot, React, Next.js

Databases & Storage: PostgreSQL, MySQL, SQLite, Supabase, MinIO (S3-compatible object storage)

AI & ML: LLM integration, RAG pipelines, vector search, embeddings, semantic search, prompt engineering, LLM evaluation, PyTorch, NumPy, sentence-transformers, Claude API

Cloud & DevOps: Docker, Git, GitHub, Vercel

Tools & Concepts: unit testing (pytest, JUnit), object-oriented design, system design, REST API design, microservices, Agile collaboration

EXPERIENCE

TAIRC (The AI Research Center)

Jun 2026 to Present

Backend and AI Systems Engineer

- Build and maintain Python backend services and REST APIs, designing relational schemas and data-access layers that support AI-enabled research and production applications.
- Integrate LLM and machine-learning components into backend services, covering model invocation, prompt orchestration, retrieval, and output evaluation.
- Convert research prototypes into deployable, tested endpoints, owning features from requirements analysis through API design, implementation, testing, and deployment.
- Work with research and engineering staff to translate technical requirements into production systems, applying software engineering fundamentals across the full development lifecycle.

LIVELY (Jillcyn Enterprises, LLC) via Rutgers MBS Externship Exchange

Jan 2026 to May 2026

Full-Stack Engineering Extern

- Co-built LIVEY, a full-stack event-discovery platform, developing React search and filtering components and engineering REST API routes across the front and back end.
- Integrated Supabase (PostgreSQL) for real-time data storage, user authentication, and event management across the platform.

Little Red Riding Hood Inc. via Rutgers MBS Externship Exchange

Sep 2025 to Dec 2025

Software Engineering Extern

- Engineered secure authentication for a safety-focused rideshare platform, implementing role-based login and credential management for separate rider and driver accounts.
- Built driver session tracking and real-time geolocation handling on cloud-deployed (AWS) services, supporting dispatch coordination and passenger-safety workflows.

KAVACH (VIANSEC Solutions Pvt. Ltd.) Hyderabad, India

Sep 2023 to Aug 2025

- Collaborated with the engineering team to integrate IoT telemetry and AI-generated detections into incident and alert workflows.
- Validated device-data flows, monitored system behavior, documented integration issues, and supported troubleshooting.

Software Engineering Intern

Sep 2023 to Aug 2024

- Contributed to dashboard functionality for user access, site operations, incident tracking, and alert management.
- Performed software testing, documented defects, and supported Docker-based deployments and application debugging.

RNR Facilities Pvt. Ltd. (now RNR Infra Pvt. Ltd.), Hyderabad, India

Aug 2022 to Aug 2023

Software and Business Systems Intern

- Contributed to development of a Java and Spring Boot work-order and vendor-tracking application backed by MySQL.
- Tested and documented application features while maintaining database records and operational reports.

PROJECTS

AgentMemry: Open-Source Memory Library for AI Agents | *Python, SQLite, sentence-transformers, pytest* pypi.org/project/agentmemry

- Built and open-sourced a ~500-line Python library that gives AI agents persistent memory through local SQLite storage and on-device embeddings, with no cloud database or API keys required, and published it to PyPI.
- Implemented semantic retrieval with cosine similarity and a Maximal Marginal Relevance (MMR) diversity mode, plus per-agent namespacing, verifying the public API with 15 passing unit tests.

Chat-with-Baavansh: AI Portfolio with Embedded RAG Assistant | *Next.js, TypeScript, Claude API* baavansh-portfolio.vercel.app

- Built a personal portfolio site with an embedded RAG assistant that answers recruiter questions from a structured knowledge base, pairing a Claude-powered engine with an automatic in-browser fallback.

TinyL Compiler | *Python, Compilers*github.com/BaavanshReddy/tiny-l-compiler

- Built a complete compiler with lexer, recursive-descent parser, AST, intermediate representation, and code generation, translating the TinyL language from a formal grammar to stack-based bytecode.

Neural Network from Scratch | *Python, NumPy, PyTorch*github.com/BaavanshReddy/neural-net-from-scratch

- Implemented a perceptron and a 3-layer neural network with hand-coded forward propagation and backpropagation, reaching 89.1% face-recognition and 86.7% digit-classification accuracy validated against a PyTorch baseline.

RuPizza: Object-Oriented Ordering Application | *Java, JavaFX, JUnit*github.com/BaavanshReddy/RUPizza

- Developed a JavaFX desktop ordering application built on object-oriented principles including inheritance, polymorphism, and the Factory pattern, featuring an order builder, store-order management, and JUnit test coverage.

Campus Event Scraper | *Python, BeautifulSoup, SQLite*github.com/BaavanshReddy/campus-event-scraper

- Built a Python web-scraping pipeline that collects and normalizes university event listings into an indexed SQLite database, staying reliable through exponential-backoff retries, URL-based deduplication, and graceful offline fallback when sources are unavailable.

RESEARCH

LLM FactCheck: Factual QA Evaluation Framework | *Python, RAG, BM25, TriviaQA*github.com/BaavanshReddy/LLM_factcheck

- Co-authored a research study benchmarking LLM factual question-answering across direct prompting, BM25 retrieval, and RAG on a controlled 100-question TriviaQA evaluation set.
- Found that BM25 surfaced supporting evidence in 90% of cases while RAG improved exact-match accuracy by only 3 points (0.66 to 0.69), isolating generation quality rather than retrieval as the bottleneck.

LEADERSHIP & ACTIVITIES

- **Community Service Officer**, Rutgers University Police Department (Jan 2025 to May 2026). Supported incident response and crowd management for large-scale university events.
- **Leadership Roles**, Alpha Phi Delta Fraternity (May 2024 to May 2026). Held three elected positions: Risk Management Chair, Head of Judiciary Board, and Co-Brotherhood Chair; led safety compliance, conduct governance, and member engagement.
- **Administration Secretary**, ENIGMA Technical Society (Aug 2023 to May 2024). Organized technical workshops and streamlined communication across society events.
- **Languages:** English, Hindi, Telugu